

A Cartesian Multipole Hierarchy for the Voxel

Closed-Form Single-Site T -Matrices of an Elastic Cube

(archived material)

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Abstract

This document is kept as a record, not as a paper. In the series it belongs to, the expansion of the field about the centre of a cell, which the hierarchy below is, has been superseded by the Legendre expansion of the field over the cell, whose single site is in closed form in Nestor [1] and whose error is that of Nestor [2].

Volume-integral (coupled-dipole) methods divide a heterogeneous elastic medium into cubic voxels, each carrying a single-site T -matrix, coupled through the Green's tensor of a background medium. The single site is usually imported: from the ellipsoid, for which Eshelby's uniformity theorem holds, or from a polarisability tuned to a lattice. We derive it instead, as the first member of a Cartesian multipole hierarchy: each voxel radiates through the Cartesian multipole moments of its source and receives the field through its gradients at its centre, the integrals over the source cube being exact at every separation, so that touching voxels and the self term, a distribution in closed form, are included and there is no radius of convergence. Its coefficients are moments of the elastodynamic Green's tensor over the cube, in closed form on four constants, and nothing in it is fitted. At the order of the uniform-strain closure a cube is a sphere plus a pure cubic anisotropy, which splits the deviatoric channel about the sphere's value in the ratio 3 : 2, and the closure departs from exact scattering by a sphere only through a real term of order $(ka)^2$, the radiation damping agreeing exactly. Its blocks gain accuracy in parity pairs, with the error of each truncation in closed form: with third gradients the single site is three hundred times more accurate than the uniform-strain closure against the exact sphere, and as a voxel scheme the hierarchy is fourth order on a smoothly graded sphere. Every result is derived symbolically and checked against an independent implementation or an exact solution.

1 Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale, and the aim of the work this paper begins is an efficient model of the wavefield in a medium of realistic complication. The models that describe such a medium are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation divides

the medium into cells and treats every cell as a scatterer: a voxel carrying a single-site T -matrix, the operator that turns the field incident on the voxel into the sources it radiates, and coupled to every other voxel through the Green's tensor of the background, with the multiple-scattering series summed.

The question: the single site. A voxel is a small region in which two functions must be represented, the medium and the wavefield, and its single-site T -matrix is the operator that turns the field incident on it into the sources it radiates. Which bases those two functions need, and what error a given choice leaves, is the subject of a companion paper [2], which isolates the error exactly on a layer of voxels and finds that every error of the scheme is made at the single site: the coupling between cubes that tile space, averaged over the source cell, is the continuum's cell integral exactly. The single site is therefore the object to get right. This paper derives it rather than importing it, and states the error of each approximation to it.

The coupled-dipole method. Replacing a heterogeneous body by a lattice of small scatterers that interact through the background Green's function is the multiple-scattering description of Foldy [3] and Lax [4]. In electromagnetics it became a numerical method as the coupled-dipole method, or discrete dipole approximation (DDA): each cell of a cubic grid is a polarisable point dipole driven by the incident field and by the fields of every other dipole [5, 6]. The method is a collocation of the volume integral equation for the electric field, with the cell's self-interaction absorbed into its polarisability [7], and Yurkin and Hoekstra [8] review it in that form. Its accuracy turns on two ingredients. This paper takes up the first for elasticity; the second is taken up in the companion paper [2]. The first is the polarisability. The Clausius–Mossotti value is the static response of a sphere; the lattice dispersion relation of Draine and Goodman [9] adjusts it so that an infinite lattice propagates waves as the continuum does to a given order in kd ; Rahmani et al. [10] instead derive it from the exact static solution for the whole body, so that the method is exact in the long-wavelength limit. The second is the coupling. Integrating the Green tensor over the source cell rather than sampling it at the centre improves the accuracy, most of all for strong contrasts [11], and the filtered coupled dipoles of Piller and Martin [12], revived by Yurkin et al. [13] for large refractive indices, replace the point dipole by a source of finite extent whose spectrum is band-limited to the grid. The coupling summed over a lattice extends the method to periodic structures [14]. Its error converges at second order in the dipole spacing when shape errors are absent [15], and for a cube, the one shape a cubic grid represents without a staircase, Yurkin and Kahnert [16] reach relative uncertainties of 10^{-7} to 10^{-3} by extrapolation in the grid. On the mathematical side, Costabel et al. [17] show that the method is a finite section of an infinite block Toeplitz matrix, stable for a large range of the contrast but not for all of it, and Costabel [18] that in a one-dimensional model it converges at second order in the interior, with a boundary layer in which the consistency error does not vanish. What this paper adds is the single site in closed form: for a cube, the polarisability is neither tuned nor imported but derived, from the moments of the elastodynamic Green's tensor over the cube, with the error of each truncation stated. In the terms of Nestor [2] the coupled-dipole method is the voxel of degree zero, and its refinements improve the polarisability and the coupling while leaving the internal field uniform.

Its elastic counterparts. The elastic problem has the same volume integral equation [19], and every cell-based solver of it is a coupled-dipole method whether or not it carries the name. Nestor [20] divided a two-and-a-half-dimensional medium into cells that interact through the background Green’s tensor, each carrying a contrast renormalised by its own near field, with the multiple-scattering series summed. Kanaun and Levin [21] discretise the inclusion problem with Gaussian approximating functions whose matrix elements are analytic. Yang et al. [22] solve a stress–velocity form of the integral equation on a grid by conjugate gradients with fast Fourier transforms, and Touhei [23] does the same for a half space with a generalised Fourier transform. Jakobsen [24] and Jakobsen et al. [25] build the seismic forward and inverse problems on the transition operator of the discretised Lippmann–Schwinger equation, the first in the acoustic approximation and the second in arbitrary anisotropic elastic media with a nine-component state vector.

This paper. It extends the single site of Nestor [20] to cubic voxels. §2 sets out the discrete medium and where its kernel comes from. §3 derives the single-site T -matrix as the first member of a Cartesian multipole hierarchy in closed form, and sets out what that gives (§3.7). §4 builds the hierarchy into a voxel scheme in three dimensions. §5 discusses the results and §6 concludes. Every result is derived symbolically in Mathematica and checked against an independent Python/Fortran implementation or an exact solution.

2 Setting

The discrete medium. Space is tiled by cubes of side d (half-side $h = d/2$). Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus $M_p = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_p = \omega/\alpha$, $k_s = \omega/\beta$. Coordinates are ordered (z, x, y) with z down. The field at a voxel is the nine-component state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

displacement followed by engineering strain. A voxel responds with a force and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting state to them, and the 9×9 lattice Green’s tensor $\mathbf{K}(i, j)$ maps a source voxel’s force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

with ψ^0 the incident state.

Where the kernel comes from. Write the medium as the background plus the contrast. With time dependence $e^{-i\omega t}$ the equation of motion is $\partial_c \sigma_{bc} + \rho \omega^2 u_b = 0$, with $\sigma_{bc} = c_{bcde} \varepsilon_{de}$ and summation over repeated indices. Splitting $\rho = \rho_0 + \Delta\rho$ and $c = c^0 + \delta c$, with $\delta c_{bcde} = \Delta\lambda \delta_{bc} \delta_{de} + \Delta\mu (\delta_{bd} \delta_{ce} + \delta_{be} \delta_{cd})$ the contrast stiffness, and moving the contrast terms to the right

leaves the background operator on the left and the contrast on the right as a body force,

$$f_b = \omega^2 \Delta \rho u_b + \partial_c m_{bc}, \quad m_{bc} = \delta c_{bcde} \varepsilon_{de}, \quad (3)$$

a force from the density contrast, and the divergence of the contrast stress m , the moment, from the stiffness contrast. With $\mathbf{G}$ the Green's tensor of the background, the displacement is $u = u^0 + \int \mathbf{G} f \, dy$, and the moment term integrates by parts,

$$\int G_{ab}(\mathbf{x} - \mathbf{y}) \frac{\partial m_{bc}}{\partial y_c} \, dy = - \int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x} - \mathbf{y}) m_{bc} \, dy = + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x} - \mathbf{y}) m_{bc} \, dy, \quad (4)$$

the boundary term vanishing because m is confined to the scatterers, and the last step holding because $\mathbf{G}$ depends only on the separation. The moment therefore radiates through the *receiver* derivative of $\mathbf{G}$, with a plus sign, when the moment is the contrast stress itself. With the moment tensor defined instead through its equivalent body force $-\partial_c[M_{bc} \delta(\mathbf{x} - \mathbf{y})]$, as in Aki and Richards [26], the same field reads $M_{bc} \partial G_{ab} / \partial y_c$, the source derivative, with $M = -m$. The two are one convention written two ways; what matters is that the derivative in the kernel and the sign in the contrast operator are chosen together. The nine-component kernel follows (Fig. 1): its force columns are $\mathbf{G}$ and its moment columns the receiver derivatives $\partial_c G_{ab}$, symmetrised over (b, c) with a factor $\frac{1}{2}$ on the shear components; its displacement rows are those entries and its strain rows their receiver derivatives, symmetrised in the same way and doubled on the shear rows to give engineering strain. In that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (5)$$

where $\Delta \mathcal{C}$ is the 6×6 Voigt form of δc , the contrast stiffness with $\Delta \lambda + 2\Delta \mu$ and $\Delta \lambda$ in the normal block and $2\Delta \mu$ (not $\Delta \mu$) on the shear diagonal.

Averaging, and collocation. Three operations in this paper are averages over a cell, and they are distinct. The *source-cell average* integrates the kernel over the cell that radiates: $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r} - \mathbf{y}) \, dy$ is the field at $\mathbf{r}$ of a uniform polarisation of the source cell. Every scheme in this paper uses it, and it is exact for cubes that tile space [2]. *Collocation* fixes where the discrete equation is imposed: at the receiving cell's centre, with the internal field of each cell uniform, so that the unknown is the field at the centre. The *receiver-cell average* imposes the equation in the mean over the receiving cell instead; with a uniform internal field that is the Galerkin scheme of degree zero, called the mean-only voxel below. The two are both *uniform-field* voxels, a term that refers to the internal field and not to the test, and they differ only in the test function (a delta at the centre against the cell's indicator) and in the constant of their $O((kd)^2)$ error [2]. The *cell-mean contrast*, finally, is an average of the material, not of the field: where the medium varies within a cell, the contrast is projected onto the cell's basis, its mean for degree $r = 0$, rather than sampled at the centre [2]. In the language of the coupled-dipole method, the point dipole is collocation with the point kernel, the integrated Green tensor adds the source-cell average, and the filtered coupled dipoles replace the point source by one of finite extent.

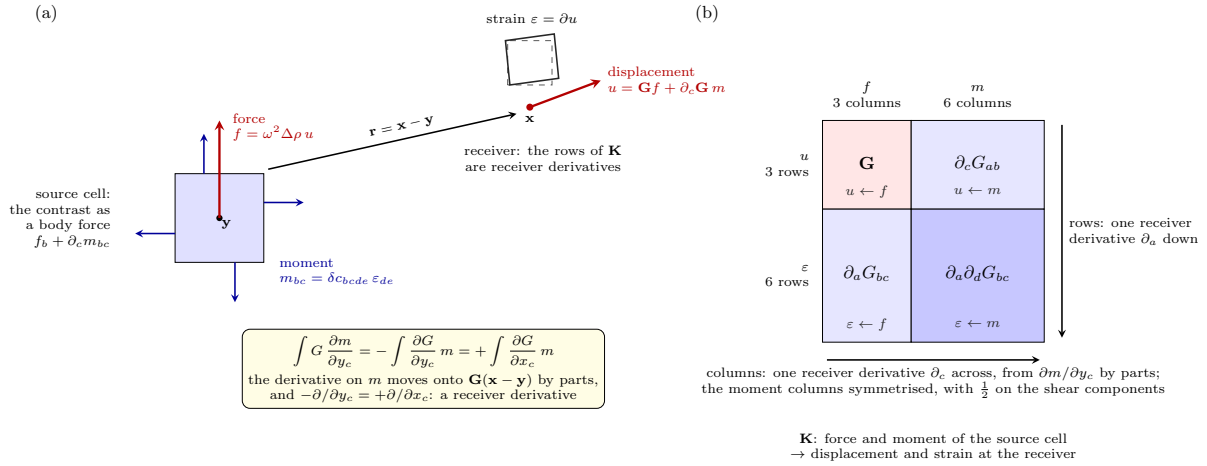


Figure 1. Where the kernel comes from. (a) A source cell at $\mathbf{y}$ acts on the background through the body force of Eq. (3): a force $\mathbf{f}$ from its density contrast, and the divergence of its contrast stress $\mathbf{m}$, the moment, from its stiffness contrast (arrows on the faces). At the receiver $\mathbf{x}$ the state is the displacement $\mathbf{u}$ and the strain $\boldsymbol{\varepsilon}$. The moment's derivative moves onto $\mathbf{G}(\mathbf{x} - \mathbf{y})$ by parts, Eq. (4), and becomes a receiver derivative with a plus sign. (b) The block structure of the 9×9 kernel $\mathbf{K}$ that results: the force columns are $\mathbf{G}$, the moment columns carry one receiver derivative across, the strain rows one receiver derivative down, and the strain-from-moment block two.

Alt text: Two panels. On the left, a shaded square cell with a dot at its centre marked $\mathbf{y}$; a thick red arrow labelled force rises from the centre, and four blue arrows labelled moment point outward from the cell's faces. A long arrow labelled $\mathbf{r}$ equals $\mathbf{x}$ minus $\mathbf{y}$ runs from the cell to a red dot marked $\mathbf{x}$ at the upper right, from which a red arrow labelled displacement leaves and above which a small dashed square deformed into a parallelogram is labelled strain. A boxed line below reads: the integral of G times the y -derivative of m equals minus the integral of the y -derivative of G times m , equals plus the integral of the x -derivative of G times m . On the right, a two-by-two grid of shaded blocks with rows labelled $\mathbf{u}$, three rows, and strain, six rows, and columns labelled $\mathbf{f}$, three columns, and $\mathbf{m}$, six columns; the blocks read $\mathbf{G}$, the first derivative of G , the first derivative of G , and the second derivative of G ; an arrow down the right side reads rows: one receiver derivative down, and an arrow along the bottom reads columns: one receiver derivative across.

Implementations. Every result is derived symbolically in Mathematica and compared with an independent numerical implementation or an exact solution. “The package” below is the Python/Fortran implementation of the voxel scheme (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified propagator, which serve both as the object under test and, where validated, as a second implementation.

3 The single site: the cube T -matrix from moment integrals

Every error of a voxel scheme is made at the single site [2], so the single site is derived here rather than imported. The classical route imports it. For an ellipsoid, Eshelby's uniformity theorem makes the internal strain under a uniform load uniform, with a depolarisation tensor in closed form [27–29]; its dynamic generalisation is known for spheres and ellipsoids [30–32]. For a cube or a general polyhedron the static field of a uniform eigenstrain is also in closed form [33–35], and it has been computed for a cuboid in an anisotropic matrix [36], but the strain it produces is not uniform, so there is no exact uniform-strain inclusion to import. Closed-form Eshelby tensors for polyhedral inclusions with polynomial eigenstrains have since been given [37], the dynamic inclusion problem has been formulated for shapes other than the ellipsoid, cubes

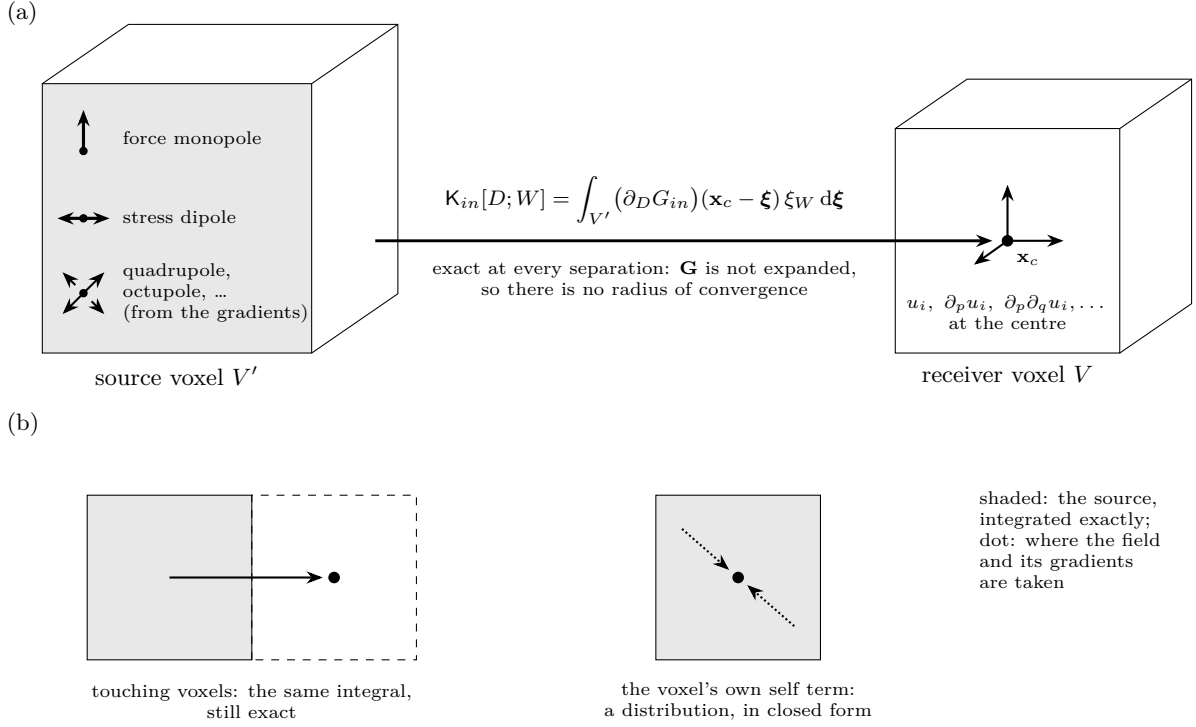
among them [38], and the Green's tensor has been integrated analytically over a cuboid for the coupled-dipole method, to an error of $O((kd)^4)$ [39]; these are the static and electromagnetic counterparts of individual moments below. In scattering theory the volume integral equation of an elastic inclusion [19] is closed in the same spirit by a self-consistent or effective-field approximation [40, 41]: a first-order (Born) scatterer [42] renormalised by the geometric series of its own near field, the step that turns a contrast into an effective contrast. Nestor [20] introduced the single site in this form, summing the series of the cell's self integral, with the singular near field evaluated over an infinitesimal cell. The moments below evaluate the same self integral over the finite cube, as a distribution. The elastic T -matrix of Waterman [43] represents a scatterer in spherical multipoles, the natural basis for a sphere; a voxel in a lattice needs one compatible with the lattice, and the moments over the cube provide it. The construction below makes the approximation of a voxel explicit rather than implicit: it derives the cube's T -matrix from the moments of the elastodynamic Green's tensor over the cube, with the uniform-strain closure as the first truncation of a hierarchy.

A Cartesian multipole hierarchy. We call the construction the Cartesian multipole hierarchy (Fig. 2). The contrast acting on the voxel's field is a source: the density contrast a body force $\omega^2 \Delta \rho \mathbf{u}$, the stiffness contrast a stress $\delta \mathbf{c} : \boldsymbol{\varepsilon}$. Each voxel radiates through the Cartesian multipole moments of these sources over the cube, a force monopole and a stress dipole from the uniform field (the cell of the coupled-dipole method) and higher multipoles from the field's gradients, and receives the field through its gradients at its centre. The name needs one caution. A multipole expansion usually expands the kernel or the field about a centre and truncates it, which holds only for well-separated cells or outside a circumscribing sphere: in the fast multipole method [44], whose elastodynamic form leaves adjacent cells to direct integration [45], in the spherical-wave T -matrix of Waterman [43], and in the force-dipole description of a point defect, an expansion of the Green's function far from it [46]. Here the Green's tensor is never expanded. Only the source is resolved into Cartesian moments, and the integral over the source cube is exact at every separation, so there is no radius of convergence, and touching voxels and the self term, a distribution in closed form, are included. The source side is shared with the multipole extensions of the coupled-dipole method in optics [47, 48] and with Cartesian multipole decompositions of a source [49]. The word hierarchy has its sense in kinetic theory and cosmology [50]: a chain of coupled moment equations closed by truncation at a chosen order, whose truncation error is here in closed form.

The multipole orders. A hierarchy of degree ℓ holds the force density $\omega^2 \Delta \rho \mathbf{u}$ to degree ℓ and the stress $\delta \mathbf{c} : \boldsymbol{\varepsilon}$ to degree $\ell - 1$. A stress moment of degree n acts on the medium through its divergence, as a force multipole of order $n + 1$, so both channels carry the force multipoles through order ℓ (Table 1). Here a multipole is the primitive Cartesian moment, which keeps its traces, rather than the irreducible one [49]. What sets the order of convergence is each channel's first omitted source term, counted in its own degree (§3.4): fourth order needs the force quadrupole for a density contrast, but the second-degree stress moments, a force octupole, for a stiffness contrast.

Table 1. The force multipoles a hierarchy of degree ℓ carries, and the order of convergence it reaches for each contrast on the layer (Table 3).

ℓ	density channel	stiffness channel	order reached	
	(force moments)	(stress moments)	density	stiffness
1	monopole, dipole	dipole	2	2
2	to the quadrupole	dipole, quadrupole	4	2
3	to the octupole	to the octupole	4	4

**Figure 2.** The Cartesian multipole hierarchy. (a) The source voxel radiates through the Cartesian multipole moments of its source: a force monopole from the density contrast, a stress dipole from the stiffness contrast, and the quadrupoles, octupoles and higher multipoles from the field's gradients. The receiver voxel takes the field through its value and gradients at its centre $\mathbf{x}_c$. The two are coupled by the integral of the derivatives of the Green's tensor over the whole source cube, Eq. (27), which is exact at every separation because the Green's tensor is not expanded. (b) The cases a multipole expansion of the Green's tensor cannot reach are therefore included: touching voxels, by the same integral, and the voxel's own self term, a distribution in closed form.

Alt text: Two panels. In the upper panel a shaded cube on the left, labelled source voxel, contains three small symbols: a single arrow labelled force monopole, a pair of opposed arrows labelled stress dipole, and a cross of four arrows labelled quadrupole, octupole, and so on, from the gradients. A long arrow runs from it to the centre of an unshaded cube on the right, labelled receiver voxel, where a dot with three short arrows marks the field and its gradients at the centre. Above the arrow is the coupling integral of the derivatives of the Green's tensor over the source cube, and below it the words exact at every separation: $\mathbf{G}$ is not expanded, so there is no radius of convergence. In the lower panel, on the left, a shaded square touches a dashed square whose centre is marked by a dot, with an arrow from the shaded square to the dot, labelled touching voxels: the same integral, still exact; on the right a single shaded square has a dot at its centre and two dotted arrows pointing to it from within, labelled the voxel's own self term: a distribution, in closed form.

Its origin in the equivalent inclusion method. In its static form the construction is the equivalent inclusion method with polynomial eigenstrains of Moschovidis and Mura [51]: the eigenstrain is expanded about the centre of the inhomogeneity, the consistency equation is expanded there in a Taylor series, and the coefficient of each monomial is equated, which amounts to imposing the Lippmann–Schwinger equation and its derivatives at the centre [52, §2.2]. Johnson et al. [53, 54] applied an expansion about the origin to a cuboidal inhomogeneity, and Wu et al. [37] expanded the eigenstrain of a polyhedron to second degree about its centroid; both report that the expansion about the centre does not follow the field near the edges, corners or vertices. Fu and Mura [30] extended the method to time-harmonic waves for an ellipsoid, with a second unknown for the density mismatch, and in the cases they report evaluated it with uniform eigenstrains. Brisard et al. [52] replaced the Taylor form by a Galerkin form of the same equation, because the Taylor form need not give an invertible system and a higher degree need not improve it. What is added here, to our knowledge, is the elastodynamic hierarchy, with density and stiffness contrasts together, to any degree and for a body that is not an ellipsoid; its moments as distributions in closed form; and the structure and the error of its truncations, tested degree by degree against the exact solution for a sphere. The moments here are integrals of the Green’s tensor against monomials over the body. They are not moments in the sense of the method of moments [55], of which the hierarchy is the instance with monomials about the centre as expansion functions and the δ function and its derivatives at the centre as testing functions; and the exact static response of an inclusion of any shape to polynomial loads is described by the polarisation and elastic moment tensors of Ammari and Kang [56].

3.1 The closed hierarchy

Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ and $\Delta\rho$. The hierarchy is stated below for any degree; its third degree is written out first, being the lowest at which the stiffness channel converges at fourth order (§3.4). Expand the internal displacement about the centre to third degree and project the Lippmann–Schwinger equation onto 1, x_p , x_px_q and $x_px_qx_v$. The unknowns are $u_i(0)$, $\partial_p u_i(0)$, $\partial_p \partial_q u_i(0)$ and $\partial_p \partial_q \partial_v u_i(0)$, and the result is the closed system

$$(\delta_{ij} - \omega^2 \Delta\rho G_{ij})u_j + (N_{in,k}^r \delta c_{nksj} - \frac{1}{2}\omega^2 \Delta\rho K_{ij}^{rs})\partial_r \partial_s u_j = u_i^0, \quad (6)$$

$$\begin{aligned} &(\delta_{pr}\delta_{ij} + \omega^2 \Delta\rho N_{ij,p}^r - M_{in,pk} \delta c_{nkrj})\partial_r u_j \\ &\quad + (\frac{1}{6}\omega^2 \Delta\rho B_{ij,p}^{rst} - \frac{1}{2}P_{in,pk}^{st} \delta c_{nkrj})\partial_r \partial_s \partial_t u_j = \partial_p u_i^0, \end{aligned} \quad (7)$$

$$-\omega^2 \Delta\rho M_{ij,pq} u_j + (\delta_{qr}\delta_{ps}\delta_{ij} + Q_{in,pqk}^r \delta c_{nksj} - \frac{1}{2}\omega^2 \Delta\rho P_{ij,pq}^{rs})\partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (8)$$

$$\begin{aligned} &(\omega^2 \Delta\rho Q_{ij,pqv}^r - X_{in,pqvk} \delta c_{nkrj})\partial_r u_j \\ &\quad + (\delta_{pr}\delta_{qs}\delta_{vt}\delta_{ij} + \frac{1}{6}\omega^2 \Delta\rho J_{ij,pqv}^{rst} - \frac{1}{2}Z_{in,pqvk}^{st} \delta c_{nkrj})\partial_r \partial_s \partial_t u_j = \partial_p \partial_q \partial_v u_i^0, \end{aligned} \quad (9)$$

all fields evaluated at the centre. Dropping the third gradients, that is Eq. (9) and the second line of Eq. (7), leaves the system of second degree; dropping the second gradients as well, that is Eq. (8) and the $\partial\partial u$ terms of Eq. (6), leaves the uniform-strain closure. Equations (6) and (8) are the same at second and third degree, because every term the third degree would add to them

vanishes by parity (§3.3). The coefficients are ten moments of the background Green's tensor $\mathbf{G}$ over the voxel

$$\begin{aligned}
 \mathbf{G}_{ij} &= \int_V G_{ij} dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r dV, \\
 \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r dV, \\
 \mathbf{B}_{ij,p}^{rst} &= \int_V \partial'_p G_{ij} x_r x_s x_t dV, & \mathbf{J}_{ij,pqv}^{rst} &= \int_V \partial'_v \partial'_q \partial'_p G_{ij} x_r x_s x_t dV, & \mathbf{X}_{in,pqvk} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} dV, \\
 \mathbf{Z}_{in,pqvk}^{rs} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} x_r x_s dV,
 \end{aligned} \tag{10}$$

the first six being those of the second degree and the last four those the third gradients add. The derivatives are taken with respect to the source point, $\partial' = -\partial$. That sign reaches the moments with an odd number of derivatives, $\mathbf{N}$, $\mathbf{Q}$, $\mathbf{B}$ and $\mathbf{J}$, and no symmetry, parity or scaling check can detect it, so it is fixed by convention. The moments are graded by (D, W) , the number of derivatives and the degree of the weight: $\mathbf{G}(0, 0)$, $\mathbf{K}(0, 2)$, $\mathbf{N}(1, 1)$, $\mathbf{M}(2, 0)$, $\mathbf{P}(2, 2)$, $\mathbf{Q}(3, 1)$, and $\mathbf{B}(1, 3)$, $\mathbf{J}(3, 3)$, $\mathbf{X}(4, 0)$, $\mathbf{Z}(4, 2)$.

The hierarchy at any degree. Nothing in the construction is special to the third degree. Let the internal displacement be expanded to degree ℓ ,

$$u_j(\mathbf{x}) = \sum_{|W| \leq \ell} \frac{x_W}{|W|!} U_{j;W}, \quad U_{j;W} = \partial_W u_j(0), \quad x_W = x_{w_1} \cdots x_{w_m}, \tag{11}$$

the sum running over ordered sets of indices $W = (w_1, \dots, w_m)$, and let the strain be its derivative. Write the moments with any number of derivatives and any weight as

$$\mathbf{g}_{in}[D; W] = \langle \partial_D G_{in}, x_W \mathbf{1}_V \rangle, \tag{12}$$

the derivatives being those of G_{in} as a function of its argument and the pairing distributional (§3.2). The ten moments of Eq. (10) are these $\mathbf{g}$ themselves: they are the integrals of $\partial'_D G_{in}(\mathbf{0} - \mathbf{x})$, and the factor $(-1)^{|D|}$ of the source derivative cancels the parity $(-1)^{|D|}$ of $\partial_D G_{in}$, so that, for example, $\mathbf{N}_{in,k}^r = \mathbf{g}_{in}[k; r]$ and $\mathbf{Z}_{in,pqvk}^{rs} = \mathbf{g}_{in}[pqvk; rs]$. Beyond the third degree the moments are written as $\mathbf{g}$ only. Imposing the Lippmann–Schwinger equation and its derivatives to order ℓ at the centre gives, for every set of derivative indices P with $|P| \leq \ell$,

$$U_{i;P} - (-1)^{|P|} \omega^2 \Delta \rho \sum_{|W| \leq \ell} \frac{\mathbf{g}_{ij}[P; W]}{|W|!} U_{j;W} + (-1)^{|P|} \sum_{|W| \leq \ell-1} \frac{\mathbf{g}_{in}[Pk; W]}{|W|!} \delta c_{nkrj} U_{j;rW} = \partial_P u_i^0(0). \tag{13}$$

Equations (6)–(9) are the case $\ell = 3$, with the moments of Eq. (10) put for the $\mathbf{g}$; their truncations above are $\ell = 2$ and the uniform-strain closure $\ell = 1$. With the field rather than the eigenstrain as the unknown, the moments $\mathbf{g}[D; W]$ play the part of the polynomial Eshelby tensors of the equivalent inclusion method [29, Ch. 4], which Brisard et al. [52] call influence pseudotensors, here for the elastodynamic kernel and for the density channel as well. For an ellipsoid the static

self term of a polynomial eigenstrain is itself a polynomial of the same degree, so the expansion about the centre is exact in the self term and approximate only in the interaction with other bodies [52]; for a cube it is approximate in both. The cube is centrosymmetric, so a moment vanishes unless $|D| + |W|$ is even, and the system needs the grades

$$\begin{aligned} (D, W) &= (m, W), \quad W \leq \ell && \text{(density),} \\ (D, W) &= (m + 1, W), \quad W \leq \ell - 1 && \text{(stiffness),} \end{aligned} \quad m = 0, \dots, \ell, \quad D + W \text{ even.} \quad (14)$$

Table 2 lists what each degree adds. The unknowns number $3\binom{\ell+3}{3}$: 12, 30, 60 for $\ell = 1, 2, 3$.

Table 2. The grades (D, W) of moments, D derivatives on the Green's tensor and a weight of degree W , that the hierarchy of degree ℓ needs beyond those of degree $\ell - 1$.

ℓ	new grades	moments in all
1	(0, 0), (1, 1), (2, 0)	3
2	(0, 2), (2, 2), (3, 1)	6: G, K, N, M, P, Q
3	(1, 3), (3, 3), (4, 0), (4, 2)	10: with B, J, X, Z
4	(0, 4), (2, 4), (4, 4), (5, 1), (5, 3)	15
5	(1, 5), (3, 5), (5, 5), (6, 0), (6, 2), (6, 4)	21

Two independent systems. In either sum of Eq. (13) the unknown $U_{i,P}$ is coupled to $U_{j;W}$, or to $U_{j;rW}$, through a moment of grade $(|P|, |W|)$ or $(|P| + 1, |W|)$, and on a centrosymmetric body that moment vanishes unless $|P|$ and the degree of the unknown it multiplies have the same parity. The hierarchy of any degree therefore separates exactly, at every frequency and contrast, into two systems: one for the even gradients $u, \partial\partial u, \dots$ and one for the odd gradients $\partial u, \partial\partial\partial u, \dots$. Raising the degree from odd to even adds a block to the even system alone, and from even to odd a block to the odd system alone. At $\ell = 3$ the even system is Eqs. (6) and (8), unchanged from $\ell = 2$, and the odd system is Eqs. (7) and (9). Of the sixty unknowns, twenty-one belong to the even system and thirty-nine to the odd. The leading radiation of a density contrast, $\int_V \Delta\rho u \, dV$, is drawn from the even system, and that of a stiffness contrast, $\int_V \delta c : \nabla u \, dV$, from the odd one. The separation is thus the structure behind the result of §3.4, that the blocks improve the density and the stiffness channels alternately. At first degree it is implicit in the dynamic equivalent inclusion method for an ellipsoid, where the force and the strain systems uncouple at the centre [30]; here it holds at every degree, for any centrosymmetric body.

3.2 The moments as distributions

Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (15)$$

[26, 57]. Expanding g_c in powers of r reduces every moment in Eq. (10) to a finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (16)$$

For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable. $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the integral depends on the shape of the region excluded around the origin. This is the source of the δ -function term of the inclusion problem, which a numerical quadrature misses. Without it the depolarisation is wrong; for a sphere it vanishes altogether (§3.3). It is not added afterwards here. The derivatives are distributional from the start, and each is moved onto the voxel's faces by pairing with the test function $w \mathbf{1}_V$, using $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$:

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (17)$$

The surface term is evaluated on the faces, a distance h from the origin, where every derivative of r^m is smooth. The recursion lowers D and W together and ends on convergent integrals. The face integrals are done in closed form by the corner sums of the tiling identity [2]. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (18)$$

which the face route satisfies exactly and a spherical excision returns as 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact. Because every power of r is paired this way, the dynamic moments keep the even-power terms of the expansion ($k^2 r, r^3, \dots$) as well as the odd ones. A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses those terms, at a real $O((kh)^2)$ cost (§3.6).

Every grade, in closed form on four constants. Closed forms for monomial-weighted integrals of $1/r$ over bricks are known [58], as are closed forms for the potential and its first two derivatives over polyhedra with polynomial density, from the gravity literature [59]. The engine used here differs in taking derivatives of any order as distributions, in treating the kernel r as well as $1/r$, and in the even-power terms of the dynamic series. The recursion (17) does not depend on the grade, so the same engine gives every scalar moment of Table 2. Each is evaluated once and stored. The scalar moments of every grade through $\ell = 5$, with up to six derivatives and weights to the fifth degree, have been evaluated for the kernels r^{-1} and r : 4224 closed forms. (The second term of Eq. (15) places two further derivatives on r ; those moments are stored through $\ell = 3$ and are evaluated numerically, by the second route below, beyond it.) Every one of them is a power of the voxel's width times

$$q_0 + q_1 \sqrt{3} + q_2 \pi + q_3 \log(2 - \sqrt{3}), \quad q_i \text{ rational}. \quad (19)$$

The integration returns many of them in other forms, with $\log(\sqrt{3} - 1)$, $\log(1 + \sqrt{3})$, inverse hyperbolic tangents and logarithms of powers of two, which are equal in value but not canonical:

two equal moments can then differ in appearance, and sums of large logarithms that cancel lose digits. Each stored form is therefore reduced to Eq. (19) by an integer-relation search at 400 digits, with $\log 2$ and $\log 3$ admitted to the search, and the reduction is accepted only if it agrees with the stored form to 1200 digits. All 4224 reduce, and the coefficients of $\log 2$ and $\log 3$ are zero in every one.

Every grade is gated by two checks whose answers do not depend on the engine, the Laplacian sum rules of Eq. (28), which test the δ function of the inclusion problem and its derivatives where a quadrature fails, and confirmed by a second route that shares nothing with the faces of the cube, to 1.1×10^{-12} or better (Appendix A).

3.3 Parity, and the nine-component block

The cube is centrosymmetric. A moment that couples a field quantity of degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and $\partial\partial u$ have degrees 0, 1 and 2, so at second degree Eq. (7) contains the first gradient alone: ∂u is determined by its own block and couples to neither u nor $\partial\partial u$. At any degree it couples to nothing of even degree; from $\ell = 3$ it couples to $\partial\partial\partial u$ (the second line of Eq. (7)). The leading scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which needs only ∂u . The leading far field of the stiffness contrast is therefore fixed by Eq. (7), and *no treatment of the second-gradient block (8) can change it*. What Eq. (8) contributes reaches the far field only through the $O((kh)^2)$ coupling in Eq. (6). That is the order, and the origin (the variation of the field across the voxel), of the collocation error on the layer, which the first-moment voxel removes [2], as does the hierarchy itself once its third-gradient block is kept (§3.4).

The symmetry of the cube. The cube is unchanged by 48 operations: the 24 rotations that carry it into itself, each with or without the inversion $\mathbf{x} \rightarrow -\mathbf{x}$. They form the group O_h . Any operator built from the cube alone commutes with all 48, and group theory then says how far it can be simplified [60, 61]. The quantities it acts on split into irreducible representations, the smallest sets of components that the symmetry operations mix only among themselves. Such an operator has no matrix elements between different representations, and on a representation that occurs once it is a multiple of the identity (Schur's lemma). It is then diagonal, with one eigenvalue for each representation, however many components that representation has. The labels are the standard ones of crystal physics. A , E and T are representations of dimension one, two and three, and the subscript g (from the German *gerade*, even) marks those that are even under inversion, as a strain is. For the nine components of a displacement gradient, A_{1g} is the dilatation, E_g the two traceless combinations of the diagonal strains, T_{2g} the three off-diagonal shears, and T_{1g} the three rotations, the antisymmetric part of the gradient. A stiffness contrast acts on strain only, so on T_{1g} the operator below is the identity. A sphere is invariant under every rotation, not only the cube's, and there E_g and T_{2g} join into a single five-dimensional shear channel. The cube splits that channel into two and three dimensions, which is why its two shear eigenvalues are averaged with the weights 2 and 3 in Eq. (23) below.

With $\Delta\rho = 0$, Eq. (7) at second degree is the 9×9 operator $\delta_{pr}\delta_{ij} - M_{in,pk}\delta c_{nkrj}$, built from

$\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (20)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (21)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (22)$$

The density channel is the displacement block of Eq. (6) with the second gradients dropped, an amplification $(1 - \omega^2 \Delta\rho \mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral of $\mathbf{G}$.

The cube against the sphere. For a sphere, Eshelby's tensor gives a dilatational channel $1 + \Delta K/(\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's dilatational channel, Eq. (20), is the sphere's exactly: its eigenvalue involves $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equation makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on the shape. The cube's shear channels differ from each other. Their isotropic average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (23)$$

The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this order is therefore a pure cubic anisotropy. It splits the deviatoric channel about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g , and the diagonal shear is the more depolarised. At the background of the tests ($\alpha = 5000$, $\beta = 3000 \text{ m s}^{-1}$, $\nu = 7/32$), $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is 0.4960.

Where Eshelby enters. Equations (20)–(22) follow from the moment integrals and the group theory alone. No ellipsoidal result is used, and no δ term is added by hand. What corresponds to Eshelby is the *truncation*. Keeping ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by the uniformity theorem that ansatz is exact if and only if the inclusion is an ellipsoid. For a sphere the construction is therefore exact in the static limit, and for a cube it is an approximation. The hierarchy (13) states that approximation as the omitted blocks, rather than as an unquantified shape correction. A polynomial expansion about the centre cannot represent the edge and corner concentrations of an isolated cube's internal field at any finite degree; Table 5 measures what each degree leaves. A voxel in a uniform layer has no such concentrations: its neighbours have the same properties, so its internal field is the continuum's, which is smooth. That is why a low-degree internal field suffices in a lattice [2] although it is only an approximation for an isolated cube.

Against the exact sphere. These statements are tested against the exact elastic Mie solution, with no Eshelby formula in between (Appendix B). Applied to a ball, the construction's dilatational and shear channels equal Mie's static concentrations to 2×10^{-10} , and the cube's dilatational channel and isotropic shear average meet the same numbers; taken as a principal value the moment has no depolarisation at all, so the whole static response is the δ term. At finite frequency the closure departs from Mie only by a real term of order $(ka)^2$ at first order in the contrast, given in closed form in Eq. (29); the radiation damping is exact. That term is the uniform-field ansatz's own error, the order the hierarchy assigns to the omitted second gradients.

3.4 What each block of the hierarchy buys

A hierarchy invites the question of what its next block is worth. The answer has a simple form, and it is not the one a count of unknowns suggests: the blocks help in pairs.

The experiment. The layer of Nestor [2] ($D = 2$ m, $\Delta\lambda = 2$ GPa, $\Delta\mu = 1$ GPa, $\Delta\rho = 100$ kg m $^{-3}$ in the background above, $\rho = 2500$ kg m $^{-3}$) at normal incidence is cut into n cells, and in each the hierarchy of degree ℓ is imposed: the displacement is its Taylor polynomial of degree ℓ about the centre, the strain the derivative of that, and the equation and its first ℓ derivatives hold at the centre, with the δ terms kept. In one dimension the moments are elementary and the exact layer is known, so the order of each truncation can be measured for a density contrast, a modulus contrast and both. The schemes are computed symbolically to 60 digits for $\ell = 1$ to 6 and, to $\ell = 3$, independently in double precision.

The prediction. A cell radiates the integral of its source against the kernel. If the source is held to a polynomial of degree d about the centre, the first power it omits is $d + 1$. When $d + 1$ is odd, that term integrates to zero against the even part of the kernel and the error is two orders smaller; when $d + 1$ is even it does not. The density source is the displacement, of degree ℓ ; the stiffness source is the strain, of degree $\ell - 1$. So a density contrast gains an order pair at every even ℓ , and a stiffness contrast at every odd ℓ . This is the parity of §3.3, read as an order of convergence.

Table 3. The order of convergence of the hierarchy of degree ℓ on the layer, measured from 16 to 32 cells in reflection and in transmission at $\omega = 300$ rad s $^{-1}$, and its reflection error on 16 cells with both contrasts. Every measured order equals the predicted one to the digits shown.

	$\ell = 1$	2	3	4	5	6
density contrast only	2	4	4	6	6	8
modulus contrast only	2	2	4	4	6	6
both	2	2	4	4	6	6
error, both, $n = 16$	5.4×10^{-6}	1.4×10^{-6}	5.8×10^{-12}	9.5×10^{-13}	2.6×10^{-18}	3.1×10^{-19}

The result. Table 3: all thirty-six orders are as predicted. The second gradients raise the density channel to fourth order and leave the stiffness channel at second, with a constant three to four times smaller in reflection; the third gradients bring the stiffness channel to fourth order;

and so on in pairs. With both contrasts, fourth order therefore needs $\ell = 3$ and sixth order $\ell = 5$. The truncation of Eqs. (6)–(9) at second gradients is one block short of the next order for a stiffness contrast, which is why it improves the uniform-strain closure by a constant and not by an order.

The error in closed form. At first order in the contrast a cell whose source is held to its Taylor polynomial of degree d returns the exact term times $1 + E_d$, where, with h the half-width,

$$E_d = \begin{cases} (-1)^{(d-1)/2} \frac{(kh)^{d+1}}{(d+2)!} & d \text{ odd, } R \text{ and } T, \\ (-1)^{d/2} \frac{(kh)^{d+2}}{(d+2)!} & d \text{ even, } R, \\ (-1)^{d/2+1} \frac{(d+1)(kh)^{d+2}}{(d+3)!} & d \text{ even, } T, \end{cases} \quad (24)$$

R and T standing for reflection and transmission, with $d = \ell$ for the density term and $d = \ell - 1$ for the stiffness term. It follows from the series of $\int_{-1}^1 P_d(ikhx) e^{\mp ikhx} dx$ with P_d the Taylor polynomial of e^{ikhx} , and it is confirmed symbolically for $d = 0$ to 6 and against the scheme on one cell, at a contrast of 10^{-20} and $kh = 0.03$, to 0.5% in all twelve cases. For $d = 0$ it is $(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission: the constants of the collocation scheme [2], which are thus the first rung of a ladder (`ContinuumLimit_GradientHierarchy.wl`, `scripts/measure_layer_taylor_hierarchy.py`).

3.5 The single site against a sphere and a cube

The hierarchy is a statement about one scatterer, so it is tested first on one scatterer. The assembly of Eq. (13), with the full tensor moments in their dynamic form (the series of e^{ikr}/r taken to r^{11}), is the same for any centrosymmetric body; only the scalar moments change.

A sphere. For a ball the scalar moments are elementary, by the surface route of Eq. (17) on $|\mathbf{r}| = a$, and the exact answer is known. A homogeneous sphere of radius $a = 10$ m with the contrast of the layer is illuminated by a P plane wave; the hierarchy's polynomial source is radiated to nine scattering angles between 11° and 169° , and the error is the largest component of the difference from the exact displacement [62, 63] over the largest exact component. The prediction, from the pairs of §3.4 with kh replaced by ka , is an error of $O((ka)^2)$ for $\ell = 1$ and 2 and of $O((ka)^4)$ for $\ell = 3$, and $O((ka)^4)$ already at $\ell = 2$ for a density contrast alone. Table 4 bears the prediction out. At $k_5 a = 0.4$ the error of the uniform-strain closure is 1.4%, that of the second-gradient system 0.32%, and that of the third-gradient system 0.0046%: three hundred times below the closure. The systems are well conditioned, the sixty equations of $\ell = 3$ having a condition number of 5.9 (`scripts/measure_ball_gradient_hierarchy.py`).

A cube. The same assembly with the cube's moments gives the single site of a voxel. No exact solution exists for a cube, so the reference is the same cube, of half-width $h = 10$ m, solved

Table 4. One homogeneous sphere by the hierarchy of degree ℓ : relative far-field error against exact scattering, and the order between the two lowest frequencies. All nine orders are as predicted.

contrast	ℓ (unknowns)	$k_S a = 0.4$	0.2	0.1	order	predicted
density only	1 (12)	6.4×10^{-3}	1.6×10^{-3}	3.9×10^{-4}	2.0	2
	2 (30)	4.8×10^{-5}	2.6×10^{-6}	1.8×10^{-7}	3.9	4
	3 (60)	1.5×10^{-5}	9.5×10^{-7}	6.9×10^{-8}	3.8	4
moduli only	1	1.9×10^{-2}	5.0×10^{-3}	1.1×10^{-3}	2.3	2
	2	5.4×10^{-3}	1.4×10^{-3}	3.4×10^{-4}	2.0	2
	3	6.2×10^{-5}	4.1×10^{-6}	2.0×10^{-7}	4.3	4
both	1	1.4×10^{-2}	3.1×10^{-3}	7.2×10^{-4}	2.1	2
	2	3.2×10^{-3}	6.9×10^{-4}	1.8×10^{-4}	1.9	2
	3	4.6×10^{-5}	2.5×10^{-6}	1.4×10^{-7}	4.2	4

by the Legendre voxels of Nestor [2] on refined grids, a cube being the one body such a grid represents exactly. Solutions with first and second moments on 2^3 and 3^3 cells differ among themselves by at most 2×10^{-5} , which is the accuracy of the reference. Table 5 shows what

Table 5. One homogeneous cube of half-width h by the hierarchy of degree ℓ : relative far-field error against the converged reference.

ℓ (unknowns)	$k_S h = 0.4$	0.2	0.1	0.05
1 (12)	2.1×10^{-2}	5.4×10^{-3}	2.8×10^{-3}	2.7×10^{-3}
2 (30)	3.2×10^{-3}	1.6×10^{-3}	1.9×10^{-3}	2.4×10^{-3}
3 (60)	3.5×10^{-4}	3.4×10^{-4}	2.9×10^{-4}	3.1×10^{-4}

distinguishes a cube from a sphere. Each truncation has a floor that does not fall with the frequency, because the internal field of an isolated cube is not a polynomial even statically: it concentrates at the edges and corners, as the static expansions about the centre of Johnson et al. [54] and Wu et al. [37] found for the cuboid and for polyhedra. The floor is 2.7×10^{-3} for the uniform-strain closure and nearly the same with second gradients, as §3.3 requires, the static stress dipole being fixed by the first-gradient block and coupled next to the third. The third-gradient block lowers it ninefold, to 3×10^{-4} . Above the floor the error falls with the frequency as on the sphere: at $k_S h = 0.4$ the third-gradient system is sixty times more accurate than the closure. The dip of the second-gradient row at $k_S h = 0.2$ is a cancellation between its static and dynamic terms, which have opposite signs. The condition numbers lie between 11 and 13 (`scripts/measure_cube_gradient_hierarchy.py`). These results come from one implementation, checked against the exact sphere and against the converged cube; the moments it uses are those confirmed by two routes in §3.2.

3.6 The collocation closure

The collocation closure takes the internal field of a voxel as uniform, the truncation of §3.3 for the density and stiffness channels together. The voxel’s own field at its centre is then $\mathbf{S}_{\text{self}} \Delta \mathbf{e}$ for internal state $\mathbf{e}$, where $\mathbf{S}_{\text{self}}$ is the nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is

block-diagonal. So $e = \psi + \mathbf{S}_{\text{self}}\Delta e$ and

$$\mathbf{T} = d^3 \Delta(\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)^{-1}. \quad (25)$$

With the source-averaged kernel of a lattice of cubes the self term cancels, and the discrete system becomes the collocation of the continuum Lippmann–Schwinger equation with a strain piecewise constant over voxels: every block of its kernel is an exact cell integral, and $\mathbf{S}_{\text{self}}$ does not appear [2].

The package’s cube T -matrix is this closure statically: both blocks agree with Eq. (25) to 2×10^{-5} at $k_{\text{sh}} = 0.01$, the difference falling as $(kh)^2$ (`ContinuumLimit_CubeT.w1`). It departs dynamically by about $0.25(kh)^2$ through its far-field form-factor corrections. Its self integrals are exact in the static and radiation parts but omit the real even-power terms of the Green’s tensor’s expansion ($k^2 r, r^3, \dots$), an error of $0.4\text{--}0.6(kh)^2$ against the analytic self term, whose series agrees with the closed-form elastodynamic tensor to 10^{-14} . Since the self term does not appear in the collocation scheme, that error cannot enter the closure scheme.

3.7 What a closed-form single-site T -matrix gives

The Cartesian multipole hierarchy costs more to set up than the alternatives: moments of the Green’s tensor to fourth derivatives, evaluated as distributions and reduced to closed form. What that buys follows from the Lippmann–Schwinger equation of one body rather than from a model fitted to it. *One derivation.* The static depolarisation, its dynamic corrections, the radiation damping and every higher block come from that one equation and one engine (§3.2), so they cannot be inconsistent with one another, and the static, spherical and low-frequency limits are results of the construction rather than conditions imposed on it. *Nothing is fitted.* Every coefficient of Eq. (13) is a moment of the background Green’s tensor over the body: there is no polarisability to choose and no depolarisation factor to calibrate, where other routes import the single site from the ellipsoid [27, 29] or close the volume integral equation with an effective field [40, 41]. As in the equivalent inclusion method [30, 51], the uniform-strain closure is the first member (§3.3); what the hierarchy adds is the closed form at finite frequency for a body that is not an ellipsoid, with the error of each truncation known. *The singular self-interaction is exact:* the δ -function content that a quadrature misses is part of the moments from the start, and the sum rule (18) and the traces (28) hold at every grade (§3.2). *The exact limits are built in:* the cube’s dilatational channel and its isotropic shear average equal the sphere’s, confirmed against the exact Mie solution to 2×10^{-10} and 10^{-8} (§3.3), and at finite frequency the closure departs from Mie only through a real term of order $(ka)^2$, the radiation damping agreeing exactly in all three channels (Eq. (29)). *The error is stated before anything is solved:* a truncation is a statement of which blocks are omitted; the orders come in the predicted parity pairs, all thirty-six on the layer and all nine on the sphere (Tables 3 and 4), and the leading error factor is in closed form, Eq. (24). *It improves block by block, on fixed constants:* raising the degree adds moments, confirmed by a second route that shares nothing with the first (§3.2), and no quadrature; against the exact sphere at $k_{\text{sa}} = 0.4$ the error falls from 1.4% for the uniform-strain closure to 0.0046% with third gradients, from a system of sixty equations with a condition number of 5.9 (Table 4).

That each degree improves on the one below is shown here for one sphere and one cube, not in general: in the Taylor form of the equivalent inclusion method, for interacting inhomogeneities, a higher degree need not improve the solution and the system need not be invertible [52, 64]. *It says what a body's shape does:* at the order of the uniform-strain closure a cube is a sphere plus a pure cubic anisotropy, which splits the deviatoric channel about the sphere's value in the ratio 3 : 2 (Eq. (23)). *It describes a scatterer, not only a cell:* the assembly is the same for every centrosymmetric body, only the scalar moments changing (§3.5), and the same moments are the self block of a voxel scheme (§4).

What it does not give. A polynomial about the centre cannot follow the edge and corner concentrations of an isolated cube's internal field, so each truncation there has a floor that does not fall with the frequency (3×10^{-4} with third gradients; Table 5). And in a lattice the Legendre voxel of Nestor [2] is the more economical in unknowns: both are fourth order on a smoothly graded sphere, the hierarchy with sixty unknowns a voxel and the first-moment voxel with thirty-six (§4). The hierarchy's case is not the number of unknowns; it is that it yields the single site itself, in closed form, with its error known.

4 The Cartesian multipole hierarchy as a voxel scheme

The Legendre voxels of Nestor [2] represent the field of a voxel by its moments over the voxel. The hierarchy of §3 represents it by its gradients at the centre. Both are polynomial, and §3.4 shows that on the layer the hierarchy of degree $\ell = 3$ is fourth order, as the first-moment voxel is. This section builds the hierarchy into a voxel scheme in three dimensions and compares the two.

The scheme. Each voxel c , centred at $\mathbf{x}_c$, carries the coefficients $U_{j;W}^c$ of Eq. (11). The equation and its derivatives to order ℓ are imposed at every centre (with $\ell = 1$ and no density contrast this is the static numerical equivalent inclusion method on cuboidal elements, with uniform eigenstrains, equivalence at the element centres and a fast Fourier transform, of Zhou et al. [65]):

$$U_{i;p}^c - \sum_{c'} \left\{ \omega^2 \Delta \rho \sum_W \frac{K_{ij}[P;W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} U_{j;W}^{c'} + \sum_W \frac{K_{in}[Pk;W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} \delta_{cnkrj} U_{j;rW}^{c'} \right\} = \partial_p u_i^0(\mathbf{x}_c), \quad (26)$$

$$K_{in}[D;W](\Delta) = \int_V (\partial_D G_{in})(\Delta - \xi) \xi_W d\xi. \quad (27)$$

The two kinds of block are of different character. For $c' = c$ the integrand is singular and the integral is a distribution: it is the moment of the hierarchy, $K_{in}[D;W](0) = (-1)^{|D|} \mathbf{g}_{in}[D;W]$, in closed form on the cube. For $c' \neq c$ the field point is the centre of another voxel, at least half a voxel from the source voxel, so the integrand is smooth and a tensor Gauss rule integrates it; the derivatives of the Green's tensor, to any order, come from the family $x^a y^b z^c r^p e^{ikr}$, which is closed under differentiation. All the singular work of the scheme is thus in the self block, where it is done once and in closed form; the Legendre voxel's self block is a double integral of the same singular kernel. The blocks depend on the offset alone, so the product of the system with a

vector is a convolution over the lattice and is done exactly by the fast Fourier transform. Where the contrast varies within a voxel it is projected there on polynomials of degree r , as in Nestor [2], and the source is the product of that polynomial and the field's: the moments then carry weights to degree $\ell + r$, and a block is a sum, over the monomials of the contrast, of blocks with the weight shifted.

Checks of the construction. The derivatives of the Green's tensor to fourth order agree with the closed forms of its value, gradient and second derivative, and with central differences beyond them, to 8×10^{-7} . A lattice of one voxel reproduces the single site of §3.5 to 10^{-14} . Only the offsets of one octant are computed, the others following from the parity of each index; a block obtained that way equals one computed directly to 10^{-15} . The iterative solution equals a dense solution to 7×10^{-11} where both can be run.

A body with corners. The cube of §3.5, at $k_s h = 0.2$, is cut into n^3 voxels. Table 6 gives the far-field error against the converged reference, with the Legendre voxels on the same body. On this body every degree of the hierarchy converges at about the same rate, an order near two

Table 6. A homogeneous cube cut into n^3 voxels, $k_s h = 0.2$: relative far-field error of the hierarchy of degree ℓ and of the Legendre voxels, with the unknowns of each voxel.

	unknowns a voxel	$n = 1$	2	3	4
hierarchy, $\ell = 1$	12	5.4×10^{-3}	1.7×10^{-3}	8.3×10^{-4}	5.0×10^{-4}
hierarchy, $\ell = 2$	30	1.6×10^{-3}	6.7×10^{-4}	3.7×10^{-4}	2.4×10^{-4}
hierarchy, $\ell = 3$	60	3.4×10^{-4}	7.0×10^{-5}	3.0×10^{-5}	1.7×10^{-5}
Legendre, mean only	9	5.4×10^{-3}	1.3×10^{-3}	5.8×10^{-4}	3.3×10^{-4}
Legendre, first moment	36	4.8×10^{-5}	6.0×10^{-6}		

between the two finest grids, and the degree buys a constant: thirty times from $\ell = 1$ to $\ell = 3$ on 64 voxels. That rate is the body's and not the scheme's, for the mean-only Legendre voxel shows it too: a homogeneous cube has edges and corners, whose fields no polynomial follows. The uniform-strain closure and the mean-only voxel agree on one cell, as they must, being the same approximation. And the first-moment Legendre voxel is far ahead: one cell of 36 unknowns is more accurate than eight voxels of the hierarchy of degree three with 480. A projection, which the Legendre voxel is, follows a field that is not polynomial better than an expansion about a point: the advantage of a Galerkin over a Taylor discretisation of the Lippmann–Schwinger equation that Brisard et al. [52] established for the static equivalent inclusion method.

A smooth medium. The comparison that bears on voxel schemes is in a medium without corners. First in one dimension: the layer of §3.4 with a contrast that varies in depth, projected in each cell to the degree of the field, against the exact graded layer. With the profile $\sin^2(\pi z/D)$ the orders from 16 to 32 cells are 1.99, 1.98 and 3.99 for $\ell = 1, 2$ and 3: a varying contrast leaves the pairs of §3.4 as they are. With the smoothstep of the graded sphere of Nestor [2] as the profile they are 1.97, 1.99 and 3.84, and on coarser grids they are erratic, the third-gradient scheme being worse than the second on four cells. The orders appear only when about four cells

cross the graded part of the profile (`scripts/measure_layer_taylor_hierarchy_graded.py`). That fixes the conditions of the three-dimensional test. That sphere has a shell 5 m wide, and a solver that reaches fourteen voxels across puts fewer than four voxels across it; so the core is reduced here from $a/2$ to $a/10$, which widens the shell to 9 m, and a second profile, $\sin^2(\pi x/2)$ in the fraction x of the shell's width measured from the surface, is run beside the smoothstep. The exact reference is the radial solution of Nestor [2] for each body. The contrast is linear in each voxel for $\ell = 3$, which suffices for fourth order [2], and of degree ℓ for $\ell = 1$ and 2.

Table 7. The graded sphere with a shell 9 m wide, $k_s a = 0.5$: relative far-field error against the exact solution for n_{sub} voxels across, with the apparent order between the two finest grids of each row. The lower block is the profile $\sin^2$.

	$n_{\text{sub}} = 4$	6	8	10	12	14	order
hierarchy, $\ell = 1$	1.5×10^{-3}	8.4×10^{-4}	4.9×10^{-4}	3.2×10^{-4}			1.94
hierarchy, $\ell = 2$	3.3×10^{-4}	2.1×10^{-4}	1.5×10^{-4}	1.0×10^{-4}			1.56
hierarchy, $\ell = 3$	2.6×10^{-4}	5.0×10^{-5}	1.7×10^{-5}	7.2×10^{-6}	3.6×10^{-6}	2.0×10^{-6}	3.80
Legendre, mean only	7.3×10^{-3}	3.2×10^{-3}	1.8×10^{-3}	1.1×10^{-3}			2.02
Legendre, first moment	2.9×10^{-4}	6.4×10^{-5}	2.2×10^{-5}	9.4×10^{-6}	4.6×10^{-6}	2.5×10^{-6}	3.92
hierarchy, $\ell = 3$	1.9×10^{-4}	3.8×10^{-5}	1.3×10^{-5}	5.2×10^{-6}	2.7×10^{-6}	1.5×10^{-6}	3.84
Legendre, first moment	2.2×10^{-4}	4.8×10^{-5}	1.6×10^{-5}	6.8×10^{-6}	3.4×10^{-6}	1.8×10^{-6}	4.02

Table 7 gives the result. The first-gradient scheme converges at second order, and is three to four times more accurate than the mean-only Legendre voxel on every grid, for twelve unknowns a voxel against nine. The second-gradient scheme is more accurate again but gains no order, as §3.4 predicts; its apparent order is still rising towards two. The third-gradient scheme converges at an apparent order between 3.7 and 4.1 from grid to grid, 3.8 over the range from six to fourteen voxels across, for both profiles; the first-moment Legendre voxel does the same, between 3.7 and 4.0 and 3.85 over the range. Neither has settled on four at fourteen voxels across, and the second profile, gentler than the first, changes neither: the two fourth-order schemes behave alike. On every grid from six voxels across the hierarchy's error is the smaller, by a fifth to a quarter. It carries sixty unknowns a voxel against thirty-six, so that for a given number of unknowns the Legendre voxel remains somewhat the more accurate; the wide margin of Table 6 was the corners.

What the hierarchy is for. The two schemes reach the same order by different closures of the same equation, and they differ in what they cost and in what they give besides. The Legendre voxel is the more economical in unknowns. The hierarchy's blocks are simpler: its self block is known in closed form on four constants (§3.2), and its coupling is a single integral of a smooth function where the Legendre voxel's is a double integral. The iteration converged in five steps at every size; the time goes into the tables of blocks, which are computed once for a grid and a frequency. Formed on the same footing, one block for each orbit of offsets under the 48 symmetries of the cube, the others following exactly, and every point value of the Green's tensor or its derivatives from the same compiled radial ladder, the hierarchy's tables are the cheaper: 0.6, 1.9 and 4.5 s for 6, 10 and 14 voxels across, against 4.0, 9.3 and 23.3 s for the first-moment Legendre voxel, with eight times fewer kernel evaluations. Both have a one-off cost besides, the hierarchy's

the smaller (0.3 s against 2.2 s): the moments of its self block depend on neither the frequency nor the cell size, and are computed once and stored (`scripts/build_cube_unit_moments.py`; `scripts/measure_hierarchy_table_cost.py`; one machine, five threads). And the hierarchy is, in addition, a single-site T -matrix in closed form, whose accuracy improves block by block (§3.5), which no voxel defined by moments over the cell provides. The lattice results of this section come from one implementation, checked against exact solutions and, where it can be, against a dense solve; they are at one frequency and on one family of bodies (`scripts/gradient_voxel_lattice.py`, `scripts/measure_lattice_gradient_hierarchy.py`, `scripts/measure_graded_sphere_gradient_hierarchy.py`, `scripts/measure_graded_sphere_gradient_hierarchy_fft.py`).

5 Discussion

What the single site is. The Cartesian multipole hierarchy of §3 turns the uniform-field single site from a modelling choice into a derived object with a stated error (§3.7). Its one departure from exact scattering by a sphere, the real $(ka)^2$ term of Eq. (29), and the collocation error on the layer [2] have the same origin, the variation of the field across the scatterer, and the same remedy, a richer internal field. In a lattice the remedy is cheap: the field a voxel must represent is the smooth continuum field, not the field of an isolated cube, with its concentrations at the edges and corners.

What it does not settle. The hierarchy beyond the first gradient is established less widely than the uniform-strain closure: its orders and its error factor on the layer are derived symbolically and checked independently to the third degree, and its moments are confirmed by two routes, but its three-dimensional results, the single site against the sphere and the cube and the voxel scheme of §4, come from one implementation, at one frequency for the voxel scheme, checked against exact solutions. There the fourth order is approached and not attained on the grids reached (3.8 at fourteen voxels across, as for the first-moment voxel), the fifth-gradient system, whose moments are in hand, has not been assembled in three dimensions, and the two fourth-order schemes have been compared in the cost of forming their tables only (§4).

6 Conclusions

The single site is derived rather than assumed. Three properties of the cube make the experiment exact: the source-cell-averaged coupling is the continuum's cell integral exactly, the moments of the Green's tensor over the cube give the single site with no polarisability to choose, and its self term cancels the kernel's, so the simplest scheme is a collocation of the continuum equation. The uniform-strain closure is the first member of a Cartesian multipole hierarchy, the elastodynamic extension of the Taylor form of the equivalent inclusion method [51] to bodies that are not ellipsoids, whose moments are in closed form on four constants. Its blocks gain accuracy in pairs, with the error of each truncation in closed form: as a single site the third-gradient system is three hundred times more accurate than the uniform-strain closure against the exact sphere, and as a voxel scheme it is fourth order, as the first-moment voxel is, on a smoothly graded sphere,

where in a lattice the Legendre voxel is the more economical. The results and the evidence for each are collected in Appendix C.

A The moment engine: gates and a second route

Two checks, whose answers do not depend on the engine, gate every grade. Because $\nabla^2 r^m = m(m+1)r^{m-2}$ and $\nabla^2(1/r) = -4\pi\delta$,

$$\sum_p E[m; ppR; W] = m(m+1) E[m-2; R; W], \quad \sum_p E[-1; ppR; W] = -4\pi(-1)^{|R|} (\partial_R x_W)(0), \quad (28)$$

for any further derivatives R and any weight W . The second is the sum rule (18) at every grade: it tests the δ function of the inclusion problem and its derivatives, which is where a quadrature fails. Both hold for every grade of Table 2 with two or more derivatives, 910 traces of each kind; the grades with fewer, for which the integrand is integrable, agree with numerical integration over the cube.

The closed forms are also confirmed by a route that shares nothing with the faces of the cube. The cube is its inscribed ball plus the remainder. On the ball the derivatives are moved onto the sphere, where the surface integrals of monomials are Gamma functions, and all the distributional content is there; on the remainder the integrand is an ordinary function, integrated radially in closed form and over directions through the six faces by Gauss–Legendre quadrature. This evaluation reproduces all 4224 stored values, the largest difference being 1.1×10^{-12} , and satisfies Eq. (28) by itself to 5×10^{-10} or better up to eight derivatives (CubeMomentCore.wl, CubeMomentHigherGrades.wl, CubeMomentStore.wl, CubeMomentCanonical.wl; scripts/crosscheck_cube_moments_ball_shell.py).

B The uniform-strain closure against the exact sphere

Against the exact sphere. These statements are tested against the exact elastic Mie solution for a sphere, with no Eshelby formula in between. The concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilatational, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation is applied to a ball of radius a , with its moment M evaluated by the surface route of Eq. (17) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie’s to 2×10^{-10} , and the density channel is 1. The cube’s closed forms meet the same numbers. Its dilatational channel is Mie’s E_0 to 2×10^{-10} , and its isotropic shear average, Eq. (23), is the depolarisation read from Mie’s E_2 to 10^{-8} . The same moment taken as a principal value, over the ball with a small sphere excluded around the origin, has zero depolarisation in every channel. The whole static response of the sphere is the δ term.

At finite frequency the closure uses the dynamic ball moments, which are in closed form. It departs from Mie at second order in ka and first order in the contrast, and the departure is closed-form too: the ball’s moments are elementary, and Mie’s low-frequency series with the

contrast kept exact gives E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_p a)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu'}, & n = 0, \\ \frac{2(k_s a)^2 + (k_p a)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(k_s a)^2 + 2(k_p a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (29)$$

with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms, the radiation damping, agree exactly in all three channels. So the closure's only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's own. That is the order the hierarchy assigns to the omitted second gradients (§3.3), and it has the same origin, the variation of the field across the scatterer, as the collocation error on the layer [2]. Eq. (29) is derived symbolically (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from the numerical Mie solve and an independent implementation of the closure to 5×10^{-4} , which is the size of the next order in the contrast.

C The evidence for each result, and where to reproduce it

Table 8 lists each result with the evidence for it; the results on the layer and the graded sphere, and their evidence, are in Nestor [2]. Every numerical value is reproducible from the Mathematica notebooks `Mathematica/ContinuumLimit_X.wl`, for X in Reference, ThinLayer, SpecularSums, StaticLocal, AveragedSums, CubeT, Chain, ErrorConstant, FourthOrder, FourthOrderError, Reduction3D, Oblique, IncidentS, Heterogeneous, SecondMoment, GradedContrast, LayerDefect and Figure (data for the figure), and their reference dumps `scripts/continuum_limit_*.py`; the moment integrals are `Mathematica/CubeMomentCore.wl` (gate `CubeMomentCoreTest.wl`), `CubeA22Block.wl`, `CubeMomentArchiveCheck.wl` and `CubeShearSplit.wl`, with the Python half of the last in `scripts/gate_cube_shear_split.py`; the stratified reference against Kennett's recursion is `scripts/gate_heterogeneous_reference_vs_kennett.py`; the voxelised spheres are `scripts/pilot_sphere_voxel_vs_mie.py` (sharp) and `scripts/pilot_graded_voxel_sphere.py` (graded, with the mean-only voxel of `cubic_scattering/graded_voxel`), the graded sphere's exact solution `Mathematica/ContinuumLimit_GradedSphere.wl` with the library `Mathematica/TakeuchiSaito.wl` and `scripts/crosscheck_graded_sphere.py`; the test against exact Mie is `scripts/gate_sphere_closure_vs_mie.py`, with the closed forms in `Mathematica/SphereClosureDynamic.wl`; the exact kernel is `cubic_scattering/cell_averaged_lattice.py`; the independent implementation of the first-moment voxel is `scripts/crosscheck_first_moment_voxel.py`, and of the voxel of any degree `scripts/crosscheck_second_moment_voxel.py`, and of the graded cell `scripts/crosscheck_graded_contrast.py`; the projection-error law and its split are measured by `scripts/measure_layer_bases.py`, in a stratified background by `scripts/measure_layer_stratified_background.py`, `scripts/measure_oblique_stratified_background.py` and `scripts/measure_oblique_two_term.py`, and the voxel of degree 2 at oblique incidence by `scripts/measure_oblique_degree.py`; the convergence figure is drawn by `scripts/plot_convergence_orders.py`; the near field of the sphere is mea-

sured by `scripts/measure_graded_sphere_near_field.py`; the moments of every grade are in `Mathematica/CubeMomentHigherGrades.wl`, `Mathematica/CubeMomentStore.wl` and `Mathematica/CubeMomentCanonical.wl`, with their independent evaluation in `scripts/crosscheck_cube_moments_ball_shell.py`; the hierarchy on the layer is `Mathematica/ContinuumLimit_GradientHierarchy.wl` and `scripts/measure_layer_taylor_hierarchy.py`, and in three dimensions `scripts/measure_ball_gradient_hierarchy.py`, `scripts/measure_cube_gradient_hierarchy.py` and `scripts/measure_graded_sphere_gradient_hierarchy_fft.py`. All are available at <https://github.com/tmnestor/MultipleScatteringCalculations>.

Table 8. The results of the paper, and the evidence for each.

Result	Evidence
Moments as distributions, Eq. (17)	sum rule (18) exact; M, Q equal an independent archive, 81/81
Channels (20)–(22)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (23)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube's bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (29)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Package T = closure statically	2×10^{-5} at $k_s h = 0.01$
Moments of every grade, Table 2, Eq. (19)	4224 closed forms on four constants, to 1200 digits; sum rules (28), 910 traces of each kind; independent route, 1.1×10^{-12}
Orders of the hierarchy, Table 3	36 orders, $\ell = 1$ –6, symbolic at 60 digits; independent to $\ell = 3$
Error factor of a truncation, Eq. (24)	symbolic, $d = 0$ –6; scheme to 0.5%, 12 cases
Single site against the sphere, Table 4	nine orders against exact scattering; 300 times the closure's accuracy
Single site of a cube, Table 5	reference converged to 2×10^{-5} ; static floor 2.7×10^{-3} to 3×10^{-4}
Hierarchy as a voxel scheme, Table 7	order 3.8, 6 to 14 voxels across, two profiles; dense solve to 7×10^{-11}

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This work began thirty years ago in the author's doctoral research at the Research School of Earth Sciences of the Australian National University, Canberra [20].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Software development. Software development followed a human-directed, AI-assisted workflow. The author designed the software originally, but only directed and validated recent improvements, which build on the author’s original Fortran implementation (the [PhD_fortran_code](#) folder of the repository given under Data availability) and an extensive library of Mathematica notebooks, developed and extended since 1992, that carry out and verify the symbolic derivations; an agentic coding assistant (Anthropic Claude, Opus 5.5) operated under human-in-the-loop supervision to implement specified changes and diagnostic tests. The design, the choice of numerical methods, and the resolution of the principal technical difficulties were the author’s. All AI-generated changes were subject to code review by the author and validated by automated tests and independent physical checks such as reciprocity, optical theorem, Hamiltonian symmetry, analytic limits and convergence.

Manuscript preparation. During the preparation of this work, the author used Claude (Anthropic) in order to search for relevant open-access research, to write the TikZ code of the diagrams and to help redraft the paper into more coherent sections for readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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